

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-aed4ce2e45d821e71.jsonl`  
- SHA-256: `0a5a4f04870647505635a8937b66234b2cfad3a4dd8f2168f0f0c648ce321ebd`  
- Source modified: `2026-05-31T04:22:06+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.011Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"olmOCR 2 releases its model weights (olmOCR-2-7B-1025), training data, and code under Apache 2.0, a permissive commercially-safe license ? making it directly usable for muneem's license constraints, unlike its top competitor Chandra OCR which ships under the restrictive OpenRAIL-M license."

****Source:**** <https://arxiv.org/html/2510.19817v1> (primary)

****Supporting quote:**** "We release our model, data and code under permissive open licenses.

[Table 1: olmOCR 2 ? Apache 2.0; Chandra OCR ? OpenRAIL-M]"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.715Z)

I'll verify this claim about olmOCR 2's licensing. Let me start by examining the primary source and searching for contradicting evidence.

3. assistant (2026-05-31T04:21:13.681Z)

I'll research this claim about olmOCR 2's licensing and the Chandra OCR comparison.

4. user (2026-05-31T04:21:19.251Z)

olmOCR 2 Licensing

****olmOCR 2's Licenses:****

According to the document, olmOCR 2 releases its model, training data, and code "under permissive open licenses." Specifically, Table 1 indicates olmOCR 2 uses an "Apache 2.0" license, noted as: "Open-source licenses: Apache 2.0¹"

****Chandra OCR:****

Yes, the document mentions Chandra OCR. According to Table 1, Chandra OCR 0.1.0 (released October 2025) uses an OpenRAIL-M license. The table footnote explains: "open licenses with usage restrictions: OpenRAIL-M³"

The document also references Chandra OCR in the Related Work section, describing it as one of several "notable examples" of "end-to-end systems" in the VLM-based OCR landscape.

5. user (2026-05-31T04:21:20.617Z)

Web search results for query: "olmOCR 2 model weights license Apache 2.0 olmOCR-2-7B-1025"

Links: [{"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}, {"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "lmstudio-community/olmOCR-2-7B-1025-GGUF · Hugging Face", "url": "https://huggingface.co/lmstudio-community/olmOCR-2-7B-1025-GGUF"}, {"title": "richardyoung/olmOCR-2-7B-1025-GGUF · Hugging Face", "url": "https://huggingface.co/richardyoung/olmOCR-2-7B-1025-GGUF"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Discussions", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8/discussions"}, {"title": "[Literature Review] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.themoonlight.io/en/review/olmocr-2-unit-test-rewards-for-document-ocr"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "allenai/olmocr-2-7b-1025 ? LM Studio", "url": "https://lmstudio.ai/models/allenai/olmocr-2-7b-1025"}, {"title": "bartowski/allenai_olmOCR-2-7B-1025-GGUF - MyGGUF Model Details", "url": "https://mygguf.com/models/bartowski_allenai_olmOCR-2-7B-1025-GGUF"}, {"title": "richardyoung/olmocr2:7b-q8/license", "url": "https://ollama.com/richardyoung/olmocr2:7b-q8/blobs/cf989ea142f2"}]

Based on the search results, I can provide you with information about the olmOCR-2-7B-1025 model and its license:

The olmOCR-2-7B-1025 model is released under permissive open licenses, and the specific license is Apache 2.0.

The model weights for olmOCR 2 are available on Hugging Face in both FP8 and full precision formats, along with the datasets and code used to train and fine-tune them. The olmOCR-2-7B-1025 model is fine-tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset and has been additionally fine-tuned using GRPO RL training to improve performance on math equations, tables, and other challenging OCR cases.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:21:23.984Z)

The arXiv source confirms the licensing claim. However, I need to verify a critical nuance: olmOCR 2 is fine-tuned from Qwen2.5-VL-7B-Instruct. Let me check the actual HuggingFace model card for the weights license, and verify the Chandra OCR license claim independently, since the claim's strength depends on the weights license being genuinely Apache 2.0 (not inheriting a base-model restriction).

7. user (2026-05-31T04:21:28.721Z)

License and Base Model Information

****License:**** `apache-2.0`

****Base Model:**** `Qwen/Qwen2.5-VL-7B-Instruct`

****Usage Restrictions:****

The model is licensed under Apache 2.0 and is "intended for research and educational use in accordance with Ai2's [Responsible Use Guidelines](https://allenai.org/responsible-use)."

There is no explicit mention of inheriting the base model's license in the provided content. The page only specifies the Apache 2.0 license for olmOCR-2-7B-1025 itself.

8. user (2026-05-31T04:21:31.788Z)

Web search results for query: "Chandra OCR datalab license OpenRAIL-M model weights"

Links: [{"title": "GitHub - seigneurcui/chandra-ocr: OCR model that handles complex tables, forms, handwriting with full layout.", "url": "https://github.com/seigneurcui/chandra-ocr"}, {"title": "GitHub - saintyun/OCR-chandra: OCR model that handles complex tables, forms, handwriting with full layout.r", "url": "https://github.com/saintyun/OCR-chandra"}, {"title": "GitHub - datalab-to/chandra: OCR model that handles complex tables, forms, handwriting with full layout. · GitHub", "url": "https://github.com/datalab-to/chandra"}, {"title": "GitHub - FurkanGozukara/chandra_OCR_Premium: OCR model that handles complex tables, forms, handwriting with full layout. · GitHub", "url": "https://github.com/FurkanGozukara/chandra_OCR_Premium"}, {"title": "chandra/MODEL_LICENSE at master · datalab-to/chandra", "url": "https://github.com/datalab-to/chandra/blob/master/MODEL_LICENSE"}, {"title": "GitHub - akanugan/chandra-ocr: OCR model that handles complex tables, forms, handwriting with full layout. · GitHub", "url": "https://github.com/akanugan/chandra-ocr"}, {"title": "datalab-to/chandra-ocr-2 · Hugging Face", "url": "https://huggingface.co/datalab-to/chandra-ocr-2"}, {"title": "GitHub - meekidsclub/chandraOCR: OCR model that handles complex tables, forms, handwriting with full layout. · GitHub", "url": "https://github.com/meekidsclub/chandraOCR"}, {"title": "GitHub - daemonXid/chandraOCR: OCR model that handles complex tables, forms, handwriting with full layout.", "url": "https://github.com/daemonXid/chandraOCR"}, {"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}]

Based on the search results, here's what I found about the Chandra OCR model from Datalab:

License Information

The code is Apache 2.0, and the model weights use a modified OpenRAIL-M license (free for research, personal use, and startups under \$2M funding/revenue, cannot be used competitively with our API).

Model Details

Chandra 2 is a state of the art OCR model from Datalab that outputs markdown, HTML, and JSON. It is highly accurate at extracting text from images and PDFs, while preserving layout information.

Commercial Licensing

To remove the OpenRAIL license requirements, or for broader commercial licensing, you can visit their pricing page.

The model weights are publicly available on Hugging Face under the datalab-to organization (such as `datalab-to/chandra-ocr-2`), and the code repositories are available on GitHub under the same organization.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:21:36.608Z)

Both halves of the claim are confirmed strongly. The olmOCR-2-7B-1025 weights are tagged `apache-2.0` on HuggingFace, and Chandra OCR's weights use a modified OpenRAIL-M license with commercial restrictions (research/personal/sub-\$2M startups only, no competitive use).

One nuance worth checking: the claim calls Chandra OCR olmOCR 2's "top competitor." This phrasing comes from the arXiv paper's framing. Let me verify the Chandra MODEL_LICENSE directly and confirm there's no concealed base-model license issue, then I have enough to rule.

10. user (2026-05-31T04:21:41.204Z)

License Analysis

****License Type:**** AI PUBS OPEN RAIL-M LICENSE (MODIFIED) - Version 0.1

Commercial Use Restrictions

Yes, this license imposes significant commercial restrictions:

****Revenue Cap:**** Companies generating over \$2 million in annual gross revenue are restricted unless use is limited to personal or research purposes.

****Funding Cap:**** Organizations that have raised over \$2 million in total equity or debt funding face the same restrictions.

****Non-Compete Clause:**** The license prohibits use if "You...provides or otherwise makes available any product or service that competes with any product or service offered by or made available by Licensor or any of its affiliates."

Key Restrictive Language

The license states users cannot use the model "for any purpose if You (your employer, or the entity you are affiliated with) generated more than two million US Dollars (\$2,000,000) in gross revenue in the prior year, except where Your Use is limited to personal use or research purposes."

****Commercial Licensing:**** The restriction notes that "Commercial and broader use licenses may be available from Licensor" at datalab.to, suggesting paid licensing is required for unrestricted commercial use by larger entities.

11. assistant (2026-05-31T04:21:53.357Z)

I have sufficient verification across multiple independent sources. Let me finalize my assessment.

****Verification summary:****

1. ****olmOCR 2 = Apache 2.0 (code, data, weights):**** Confirmed at three levels ? the arXiv paper Table 1 ("Apache 2.0"), the HuggingFace model card metadata (`license: apache-2.0`), and the search corroboration. The weights tag is genuinely `apache-2.0`, fine-tuned from Qwen2.5-VL-7B-Instruct (itself Apache 2.0 at the 7B size, so no inherited restriction). The only caveat is a soft "intended for research/educational use per Ai2 Responsible Use Guidelines" note, but the binding license field is Apache 2.0 ? this matches the claim.

2. ****Chandra OCR = restrictive OpenRAIL-M:**** Confirmed via the actual `MODEL\_LICENSE` file at datalab-to/chandra ? it's a *modified* AI PUBS OPEN RAIL-M with a \$2M revenue/funding cap and an explicit non-compete clause. This is materially more restrictive than vanilla OpenRAIL-M, and clearly not commercially safe under muneem's MIT/BSD/Apache requirement. The claim's characterization ("restrictive OpenRAIL-M") is accurate, if slightly understated (it's a hardened variant).

3. ****"Top competitor" framing:**** The paper does position Chandra among the leading competing

end-to-end VLM-OCR systems. This is reasonable paper framing, not a load-bearing factual error.

4. **Currency:** olmOCR-2-7B-1025 (Oct 2025) and the arXiv paper (v1, Oct 2025) are current as of May 2026. Not outdated.

5. **Source quality:** Primary sources throughout ? official HuggingFace model cards (the authoritative license source) and the original GitHub license file, not marketing summaries.

The claim is well-supported, current, and backed by primary sources. The only minor imprecision is that Chandra's license is a *modified/hardened* OpenRAIL-M (worse than the claim implies), which strengthens rather than weakens the claim's core point that olmOCR 2 is the license-safe choice. This does not refute it.

12. user (2026-05-31T04:22:03.338Z)

Structured output provided successfully

13. assistant (2026-05-31T04:22:06.874Z)

Verification complete. The claim is **not refuted** ? it holds up under adversarial scrutiny with high confidence, backed by primary sources (HuggingFace model cards and the original GitHub license file). Both the olmOCR 2 Apache-2.0 weights claim and the Chandra OCR restrictive-license contrast are accurate; the only imprecision (Chandra's license is a hardened/modified OpenRAIL-M with \$2M caps plus a non-compete) strengthens rather than weakens the claim.